000029.jpg

Original caption

a blue electronic calculator

Scene graph

SCENE GRAPH (1 objects detected):
1. calculator [center-mid] conf=0.62 depth=0.340 depth_rank=1 area_rank=1
   Limited clear space immediately around the calculator.

Scene Type A — Grounded spatial description

Based on the image, a blue electronic calculator is the central and sole object in the scene. It is positioned diagonally across the center of the frame, with the top-left corner higher than the bottom-right. The calculator is shown against a plain white background, allowing for clear space around its perimeter.

Scene Type B — Grounded referring expressions

Object: calculator
Referring: The blue electronic calculator is centrally positioned and angled slightly within the frame.
Free space: There is limited clear space immediately around the calculator, which takes up a large portion of the image.

000030.jpg

Original caption

mens sued shoes with velcro strap

Scene graph

SCENE GRAPH (1 objects detected):
1. shoe [center-near] conf=0.72 depth=0.312 depth_rank=1 area_rank=1
   Limited clear space immediately around the shoe.

Scene Type B — Grounded referring expressions

Object: shoe
Referring: The blue suede shoe with black soles and two velcro straps, positioned centrally in the image.
Free space: The shoe is surrounded by clear, white space on all sides, providing ample room for manipulation without nearby obstructions.

Scene Type C — Grounded action caption

My target object is the single blue suede shoe, positioned [center-near] at a depth of 0.312. There are no other objects to obstruct the approach as it is the primary item (depth_rank=1), but there is limited clear space immediately around it. To ensure a stable grasp while respecting the constrained area, I will execute a top-down approach to secure the main body of the shoe. Given the lack of surrounding free space, I will transfer the object to an external, unobstructed placement zone.

000003.jpg

Original caption

the brain coffee mug

Scene graph

SCENE GRAPH (3 objects detected):
1. coffee cup [center-near] conf=0.72 depth=0.279 depth_rank=3 area_rank=1
   Limited clear space immediately around the coffee cup.
2. man [center-near] conf=0.82 depth=0.274 depth_rank=1 area_rank=3
   Clear space right of the man.
3. mug [center-near] conf=0.33 depth=0.279 depth_rank=2 area_rank=2
   Limited clear space immediately around the mug.

Scene Type A — Grounded spatial description

The image features a single white ceramic mug, which is identified in the scene graph as both "mug" and "coffee cup," referencing the same object. The third detected object, "man," is actually a black graphic printed directly onto the front surface of the mug, rather than a separate physical entity. The man graphic is positioned centrally on the front face of the mug, with the mug's handle visible on the right side.

Scene Type B — Grounded referring expressions

Object: coffee cup
Referring: The large white ceramic coffee cup is centered in the image and features a black graphic of a man and "THE BRAINS" text.
Free space: There is limited clear space around the edges of the coffee cup, especially at the top and left side where it extends close to the image boundaries.

Object: man
Referring: The man is depicted as a black silhouette graphic on the side of the coffee mug, walking forward in a suit.
Free space: There is a significant amount of clear white space on the surface of the mug to the right of the man graphic and below the handle.

Object: mug
Referring: The mug is the main object in the image, characterized by its large size, white color, and prominent handle on the right side.
Free space: There is limited clear space visible around the mug in the image frame, particularly above and to the left of its body.

000010.jpg

Original caption

a maroon jumper that has a small patch in the front and long sleeves in the back

Scene graph

SCENE GRAPH (4 objects detected):
1. hang hanger [center-mid] conf=0.54 depth=0.371 depth_rank=4 area_rank=4
   Clear space right of the hang hanger.
2. blouse sweater sweatshirt [center-near] conf=0.40 depth=0.192 depth_rank=3 area_rank=3
   Limited clear space immediately around the blouse sweater sweatshirt.
3. blouse sweater sweatshirt [center-near] conf=0.38 depth=0.192 depth_rank=1 area_rank=1
   Limited clear space immediately around the blouse sweater sweatshirt.
4. blouse shirt [center-near] conf=0.32 depth=0.192 depth_rank=2 area_rank=2
   Limited clear space immediately around the blouse shirt.

Scene Type B — Grounded referring expressions

Object: hang hanger
Referring: The wooden hanger located at the top of the image.
Free space: There is clear space to the left and right sides of the hanger.

Object: blouse sweater sweatshirt
Referring: The large maroon long-sleeved top hanging in the center of the image.
Free space: Limited clear space is available immediately around the edges of the garment.

Object: blouse sweater sweatshirt
Referring: The maroon top with a wide scoop neckline.
Free space: The garment occupies most of the central portion of the frame, leaving limited clear space immediately around it.

Object: blouse shirt
Referring: The dark red garment with a small pocket detail on the right side.
Free space: Clear space is available on the left, right, and bottom edges of the garment.

Scene Type C — Grounded action caption

My perception system identifies the maroon garment (blouse/sweater, objects 2, 3, 4) as the primary target in the immediate foreground (depth 0.192, depth ranks 1-3). The hanger (object 1) is positioned further back (depth 0.371, depth rank 4), supporting the garment. To grasp the garment, a frontal approach is optimal, targeting the central mass (objects 2 and 3). The limited clear space immediately

around the garment requires a precise action, but the hanger behind it does not obstruct access to the front surface. For placement or repositioning, a clear zone to the right of the hanger is available based on spatial analysis.

000022.jpg

Original caption

led floodlight with diffusers

Scene graph

SCENE GRAPH (4 objects detected):
1. wall lamp [center-mid] conf=0.50 depth=0.366 depth_rank=1 area_rank=2
   Limited clear space immediately around the wall lamp.
2. ceiling [center-far] conf=0.39 depth=0.862 depth_rank=4 area_rank=1
   Limited clear space immediately around the ceiling.
3. wall lamp [center-mid] conf=0.42 depth=0.366 depth_rank=2 area_rank=3
   Limited clear space immediately around the wall lamp.
4. fixture light fixture [center-mid] conf=0.35 depth=0.366 depth_rank=3 area_rank=4
   Limited clear space immediately around the fixture light fixture.

Scene Type B — Grounded referring expressions

Object: wall lamp
Referring: The gray and white light fixture located in the center of the image.
Free space: There is a wide, clear space surrounding the light fixture on all sides.

Object: ceiling
Referring: The white background area that spans the upper portion of the image behind the light fixture.
Free space: This area is entirely clear, with no other objects present except the light fixture in the foreground.

Object: wall lamp
Referring: The central object in the image, characterized by a square shape, gray fins on top, and a white patterned diffuser.
Free space: The light fixture is isolated against an empty background, providing ample clear space around it.

Object: fixture light fixture
Referring: The square light fixture prominently displayed in the middle of the frame, with a gray heat sink and a translucent white cover.
Free space: The space around the fixture is completely open and unobstructed by other items.

000018.jpg

Original caption

rustic wood and leather address business card

Scene graph

SCENE GRAPH (7 objects detected):
1. sign [center-far] conf=0.42 depth=0.805 depth_rank=1 area_rank=7
   Clear space below of the sign.
2. sign [center-far] conf=0.46 depth=0.819 depth_rank=4 area_rank=4
   Limited clear space immediately around the sign.
3. plaque sign [center-far] conf=0.36 depth=0.819 depth_rank=5 area_rank=1
   Limited clear space immediately around the plaque sign.
4. plaque [center-far] conf=0.38 depth=0.819 depth_rank=6 area_rank=2
   Limited clear space immediately around the plaque.
5. sign sign [center-far] conf=0.36 depth=0.819 depth_rank=3 area_rank=5
   Limited clear space immediately around the sign sign.
6. sign [center-far] conf=0.35 depth=0.805 depth_rank=2 area_rank=6
   Limited clear space immediately around the sign.
7. sign [center-far] conf=0.32 depth=0.819 depth_rank=7 area_rank=3
   Limited clear space immediately around the sign.

Scene Type A — Grounded spatial description

Based on the scene graph and image content, a single central object, described by a cluster of overlapping detections including "plaque sign" (3), "plaque" (4), and various instances of "sign" (1, 2, 5, 6, 7), dominates the scene. This single object, resembling a rustic business card or plaque, is positioned directly in the center of the frame and appears to be fastened to a wooden background. The multiple detections all refer to this one "sign" or "plaque sign," which occupies the majority of the image area in the center-far portion. There is limited clear space immediately surrounding the object.

Scene Type B — Grounded referring expressions

Object: sign
Referring: The light-colored, rustic business card in the center of the image with the text "JOHN Q PUBLIC".
Free space: There is limited clear space immediately surrounding the business card, consisting of the wooden background.

Object: sign
Referring: The business card designed to look like a rustic plaque, positioned in the center of the

image.
Free space: The space immediately around the sign is limited, consisting of the textured wooden background.

Object: plaque sign
Referring: The central object resembling a plaque with text, "JOHN Q PUBLIC", and four bolts in the corners.
Free space: The plaque sign is tightly framed with limited space around its edges, which consists of the wooden texture background.

Object: plaque
Referring: The rectangular plaque-like business card in the middle of the image.
Free space: There is limited free space around the edges of the plaque, showing the wooden background.

Object: sign sign
Referring: The central sign with a light-colored, textured surface and dark text.
Free space: The sign is centered, with limited clear space immediately surrounding its perimeter.

Object: sign
Referring: The rustic business card in the center of the image with text 'JOHN Q PUBLIC'.
Free space: The free space immediately surrounding the sign is limited by the edges of the image frame.

Object: sign
Referring: The light-colored sign that resembles a piece of parchment attached to wood.
Free space: Limited clear space exists around the sign, consisting of the wooden background texture.

000016.jpg

Original caption

custom seat covers sued for chevrolet c350

Scene graph

SCENE GRAPH (3 objects detected):
1. office chair swivel chair [left-mid] conf=0.34 depth=0.615 depth_rank=3 area_rank=2
   Limited clear space immediately around the office chair swivel chair.
2. footrest [right-near] conf=0.32 depth=0.245 depth_rank=1 area_rank=3
   Limited clear space immediately around the footrest.
3. chair swivel chair [right-near] conf=0.32 depth=0.325 depth_rank=2 area_rank=1
   Limited clear space immediately around the chair swivel chair.

Scene Type A — Grounded spatial description

Based on the visual evidence, the scene graph's object identifications are inaccurate. The image displays two black automotive seats, not office swivel chairs. The left object is an automotive seat, positioned on the left side of the frame. The right object, also an automotive seat, is located to its right. Between the two seats is a central console assembly, a part of which is labeled as a "footrest" in the scene graph. The right seat and the central console appear slightly closer to the viewer compared to the left seat.

Scene Type B — Grounded referring expressions

Object: office chair swivel chair
Referring: The black car seat located on the left side of the image.
Free space: There is ample clear space surrounding the seat against the white background, though its placement is close to the center console and the right seat.

Object: footrest
Referring: The lower portion of the center console situated between the two car seats.
Free space: This object has clear space around it against the white background, but it is tightly positioned between the two seats.

Object: chair swivel chair
Referring: The black car seat positioned on the right side of the image.
Free space: There is ample clear space around the seat against the white background, with a center console placed immediately to its left.

000009.jpg

Original caption

a model wears an elaborate creation of white paper at the berlin fashion week on march 5, 2009 in

Scene graph

SCENE GRAPH (5 objects detected):
1. goggles [center-mid] conf=0.79 depth=0.418 depth_rank=3 area_rank=5
   Clear space left of the goggles.
2. fashion show [center-far] conf=0.63 depth=0.862 depth_rank=5 area_rank=1
   Limited clear space immediately around the fashion show.
3. headdress [center-mid] conf=0.40 depth=0.437 depth_rank=4 area_rank=3
   Limited clear space immediately around the headdress.
4. costume dress [center-mid] conf=0.40 depth=0.348 depth_rank=1 area_rank=2
   Limited clear space immediately around the costume dress.
5. headdress [center-mid] conf=0.32 depth=0.389 depth_rank=2 area_rank=4
   Limited clear space immediately around the headdress.

Scene Type A — Grounded spatial description

A model, centered in the midground of the image, wears an elaborate costume. The primary costume elements include a complex, multi-component headdress worn on the model's head, and a costume dress covering the torso and lower body. A pair of goggles or eye-mask is situated on the model's face, positioned below the headdress. The entire ensemble is presented as part of a fashion show, which occupies the background space behind the model.

Scene Type B — Grounded referring expressions

Object: goggles
Referring: The small, silver, mask-like goggles covering the model's eyes.
Free space: Free space immediately around the goggles is limited by the model's face and the surrounding headdress components.

Object: fashion show
Referring: The overall environment or scene where the model is standing on a dark runway.
Free space: There is significant free space in the dark background surrounding the model and costume.

Object: headdress (first instance)
Referring: The large, elaborate white headdress on top of the model's head, featuring a structure resembling a sailing ship.
Free space: There is ample clear space in the dark background around the upper portion of the

headdress.

Object: costume dress
Referring: The model's costume, consisting of the shiny silver top and large, ruffled white skirt.
Free space: There is clear space around the large white skirt, with the dark background providing separation.

Object: headdress (second instance)
Referring: The white, stringy, tassel-like components of the headdress that hang down around the model's face.
Free space: Free space immediately around these components is limited due to their intricate arrangement and proximity to the model's face and shoulders.

000015.jpg

Original caption

the pendleton blazer with button

Scene graph

SCENE GRAPH (4 objects detected):
1. close - up [center-far] conf=0.39 depth=0.697 depth_rank=4 area_rank=1
   Limited clear space immediately around the close - up.
2. button [center-mid] conf=0.54 depth=0.485 depth_rank=2 area_rank=4
   Clear space left, above of the button.
3. blanket [center-far] conf=0.31 depth=0.681 depth_rank=3 area_rank=2
   Limited clear space immediately around the blanket.
4. close - up [center-mid] conf=0.41 depth=0.430 depth_rank=1 area_rank=3
   Limited clear space immediately around the close - up.

Scene Type A — Grounded spatial description

A close-up view focuses on a textured fabric, which covers most of the scene and is identified as a "blanket" in the scene graph. A single "button" is situated in the center-mid portion of the image, placed directly on the surface of the blanket fabric. The overall perspective is an extreme close-up, as indicated by the multiple "close - up" entries in the scene graph, emphasizing the detail of the button and the woven pattern of the underlying blanket.

Scene Type B — Grounded referring expressions

Object: close - up
Referring: The general view of the entire scene, which consists of a close-up shot of patterned fabric.
Free space: The close-up view fills nearly the entire frame, with limited free space around the edges.

Object: button
Referring: The single dark brown, round button located slightly to the left of the image center.
Free space: The button is immediately surrounded by the patterned fabric, with fabric visible to its left and above.

Object: blanket
Referring: The woven fabric with horizontal stripes of white, brown, blue, and yellow patterns that serves as the background for the button.
Free space: The blanket (fabric) covers the majority of the image, leaving limited clear space where it extends beyond the image frame.

Object: close - up
Referring: The focused area in the center of the image, highlighting the button and the surrounding fabric texture.
Free space: The central close-up area is fully occupied by the button and fabric, leaving limited clear space within this region.

Scene Type C — Grounded action caption

Based on the scene data, my primary target is the button (depth rank 2, depth 0.485), which is located on top of the fabric surface (depth rank 1, depth 0.430). To ensure a successful grasp and avoid disturbing the underlying material, the optimal approach direction for the button is from above or to the left, leveraging the identified clear space in those areas. The surrounding fabric, represented by the blanket and other close-up areas (depth ranks 3 and 4), has limited clear space, requiring a precise vertical or side approach. The most viable placement zone, given the constraints, is immediately to the left and above the button's current position on the fabric surface.

000033.jpg

Original caption

an image of a women's black leather watch with a white dial and crystals

Scene graph

SCENE GRAPH (2 objects detected):
1. watch [center-near] conf=0.54 depth=0.176 depth_rank=1 area_rank=1
   Limited clear space immediately around the watch.
2. table [center-mid] conf=0.58 depth=0.571 depth_rank=2 area_rank=2
   Limited clear space immediately around the table.

Scene Type A — Grounded spatial description

Based on the scene graph and visual information, the watch is the primary object, positioned centrally and close to the camera in the foreground. The watch rests on a table surface which occupies the background, located behind the watch.

Scene Type B — Grounded referring expressions

Object: watch
Referring: The black watch with a white dial and silver bezel studded with crystals, positioned prominently in the center foreground.
Free space: The watch takes up most of the central image space, leaving limited clear space immediately around its casing and straps.

Object: table
Referring: The wooden surface visible underneath and around the watch.
Free space: The clear space on the table surface is significantly limited by the large watch placed on top of it.

000021.jpg

Original caption

an aqua and gold lamp sitting in a room

Scene graph

SCENE GRAPH (9 objects detected):
1. lamp [center-mid] conf=0.64 depth=0.333 depth_rank=2 area_rank=6
   Limited clear space immediately around the lamp.
2. lamp shade [center-near] conf=0.60 depth=0.313 depth_rank=1 area_rank=7
   Limited clear space immediately around the lamp shade.
3. greenery [center-mid] conf=0.67 depth=0.624 depth_rank=6 area_rank=5
   Limited clear space immediately around the greenery.
4. room [center-mid] conf=0.64 depth=0.592 depth_rank=5 area_rank=2
   Limited clear space immediately around the room.
5. mantle [center-mid] conf=0.50 depth=0.435 depth_rank=3 area_rank=4
   Limited clear space immediately around the mantle.
6. window [center-mid] conf=0.70 depth=0.661 depth_rank=8 area_rank=1
   Limited clear space immediately around the window.
7. mantle window sill [center-mid] conf=0.40 depth=0.435 depth_rank=4 area_rank=3
   Limited clear space immediately around the mantle window sill.
8. wreath [left-far] conf=0.33 depth=0.734 depth_rank=9 area_rank=9
   Clear space above, below of the wreath.
9. window sill [center-mid] conf=0.33 depth=0.656 depth_rank=7 area_rank=8
   Limited clear space immediately around the window sill.

Scene Type A — Grounded spatial description

The scene centers on a decorative aqua and gold `lamp` (1) with a matching patterned `lamp shade` (2), positioned in front of a large `window` (6). The `lamp` rests on a white `window sill` (9) that extends across the lower portion of the frame. The scene graph labels `mantle` (5) and `mantle window sill` (7) appear to be redundant references to this same `window sill` surface. Above the `window` frame, a string of `greenery` (3), or garland, is draped. A separate `wreath` (8) is not visible and appears to be a misidentification of a section of the `greenery`. The entire setting is contained within a `room` (4).

Scene Type B — Grounded referring expressions

Object: lamp
Referring: The aqua-colored lamp with gold accents, located in the center of the window sill.
Free space: The lamp has limited clear space immediately around its base on the window sill due to

other decorative objects, but ample open space directly behind it.

Object: lamp shade
Referring: The tall, cylindrical, aqua lamp shade with a pattern, positioned at the top of the central lamp.
Free space: The lamp shade has clear space around it, constrained only by the window frame and wall above and the greenery garland in front.

Object: greenery
Referring: The dark green garland draped across the top frame of the window.
Free space: The greenery garland has limited clear space immediately around it as it lies flat against the wall and window frame.

Object: room
Referring: The general environment shown in the image, featuring a window and a wall.
Free space: The room contains numerous objects on surfaces, but also large areas of open space.

Object: mantle
Referring: The white, horizontal surface below the window where objects are placed.
Free space: The mantle surface has limited clear space due to the presence of the lamp and other decorations.

Object: window
Referring: The large, backlit window in the background, visible behind the lamp.
Free space: The window itself is constrained by the wall and sill structure, with objects positioned directly in front of it.

Object: mantle window sill
Referring: The white ledge below the window, serving as the surface for the lamp and decorations.
Free space: The surface of the mantle window sill has limited clear space due to the placement of decorative items.

Object: wreath
Referring: The left-most section of the greenery garland draped along the top of the window frame.
Free space: This section of greenery has limited clear space as it rests against the wall and window frame.

Object: window sill
Referring: The white ledge below the window where the decorative objects are resting.
Free space: The surface of the window sill has limited clear space due to the placement of decorative objects.

000006.jpg

Original caption

the blue sedan is parked on a street

Scene graph

SCENE GRAPH (7 objects detected):
1. sedan [center-near] conf=0.58 depth=0.181 depth_rank=2 area_rank=1
   Limited clear space immediately around the sedan.
2. curb [right-near] conf=0.48 depth=0.243 depth_rank=3 area_rank=5
   Clear space above of the curb.
3. car [right-far] conf=0.48 depth=0.912 depth_rank=7 area_rank=6
   Clear space below of the car.
4. pavement [center-near] conf=0.49 depth=0.291 depth_rank=4 area_rank=2
   Limited clear space immediately around the pavement.
5. car sedan [left-far] conf=0.46 depth=0.781 depth_rank=5 area_rank=4
   Limited clear space immediately around the car sedan.
6. license plate [left-near] conf=0.78 depth=0.161 depth_rank=1 area_rank=7
   Clear space above, below of the license plate.
7. building [center-far] conf=0.42 depth=0.882 depth_rank=6 area_rank=3
   Limited clear space immediately around the building.

Scene Type A — Grounded spatial description

Based on the scene graph and image, a blue sedan dominates the center-near foreground of the street scene. A license plate is clearly visible on the front bumper, positioned on the left side of the vehicle. The pavement forms the ground surface beneath the blue sedan and extends toward the right, where a curb runs along the edge of the street. Further back, a silver car (identified as a car sedan in the scene graph, but appearing to be an SUV or crossover) is parked to the left of the blue sedan. The building forms the central-far background, situated behind the silver car. To the far right in the background, a small portion of a red car is also visible.

Scene Type B — Grounded referring expressions

Object: sedan
Referring: The large, blue sedan that dominates the center foreground of the image.
Free space: There is limited clear space immediately surrounding the blue sedan due to its large size and position in the frame.

Object: curb
Referring: The grey concrete curb visible along the right side of the image, running parallel to the

street.
Free space: There is clear space on the pavement below the curb and clear space for a sidewalk area above the curb.

Object: car
Referring: The small red car partially visible in the far right background.
Free space: There is clear space (street surface) visible immediately below this red car in the image frame.

Object: pavement
Referring: The dark grey asphalt street pavement beneath the blue sedan.
Free space: Most of the visible pavement area is occupied by the large blue sedan.

Object: car sedan
Referring: The silver SUV (labeled car sedan) parked in the background behind the blue sedan on the left.
Free space: There is limited clear space immediately around this silver vehicle, as it is parked close to the blue sedan and the building.

Object: license plate
Referring: The white license plate attached to the front bumper of the blue sedan.
Free space: The license plate is mounted on the car, with clear space on the bumper and grille immediately surrounding it.

Object: building
Referring: The light-colored building structure in the background behind the parked cars.
Free space: There is limited clear space immediately around the building, as its edges extend beyond the image frame.

000026.jpg

Original caption

two silver bells with ribbons and ribbons

Scene graph

SCENE GRAPH (4 objects detected):
1. package [center-near] conf=0.69 depth=0.319 depth_rank=1 area_rank=1
   Limited clear space immediately around the package.
2. heart silver [right-mid] conf=0.31 depth=0.553 depth_rank=3 area_rank=3
   Limited clear space immediately around the heart silver.
3. shape [right-mid] conf=0.32 depth=0.553 depth_rank=4 area_rank=2
   Limited clear space immediately around the shape.
4. silver [left-mid] conf=0.35 depth=0.547 depth_rank=2 area_rank=4
   Limited clear space immediately around the silver.

Scene Type A — Grounded spatial description

Based on the image, a large clear package of silver bells is positioned in the center, occupying the lower half of the frame. Above this central package, two individual silver bells with heart-shaped handles and ribbons are displayed. The bell on the left, labeled `silver` in the scene graph, is located in the upper-left area. The bell on the right, labeled `heart silver` and `shape`, is located in the upper-right area. The labels `heart silver` and `shape` both appear to refer to the same individual bell on the right. The two individual bells are positioned significantly higher in the frame relative to the package.

Scene Type B — Grounded referring expressions

Object: package
Referring: The large clear plastic package containing multiple silver bells, located in the lower half of the image.
Free space: Clear space around the package is limited by the edges of the image and the individual bells displayed above it.

Object: heart silver
Referring: The individual silver bell with a heart-shaped handle located in the upper right area of the image.
Free space: There is limited clear space surrounding this bell, primarily constrained by the image boundaries and the package below.

Object: shape

Referring: The silver-colored object in the shape of a bell, located in the upper right portion of the image above the large package.
Free space: There is limited clear space immediately around this object, with the package located directly below it.

Object: silver
Referring: The individual silver bell with a heart-shaped handle located in the upper left area of the image.
Free space: There is limited clear space surrounding this bell, constrained by the image boundaries and the package below it.

Scene Type C — Grounded action caption

I see a close-up arrangement of items where the primary object, a package containing silver bells (depth rank 1, center-near), dominates the foreground. Two individual silver bells (`silver`, `heart silver`, `shape`) are positioned further back (depth ranks 2-4). To grasp the loose bells, a vertical top-down approach is necessary to secure the heart-shaped handles and avoid collision with the bell body. For the central package, a side approach targeting its larger surface area would be stable. Given the "limited clear space immediately around" all detected objects, precise movements are critical to avoid interference. The clearest placement zone for either item would be the open area in the immediate foreground, away from the current cluster.

000011.jpg

Original caption

two male basketball players sit in the middle of a crowd

Scene graph

SCENE GRAPH (9 objects detected):
1. basketball coach man [left-mid] conf=0.41 depth=0.618 depth_rank=4 area_rank=4
   Limited clear space immediately around the basketball coach man.
2. basketball team [center-far] conf=0.40 depth=0.977 depth_rank=7 area_rank=1
   Limited clear space immediately around the basketball team.
3. basketball game [center-far] conf=0.38 depth=0.977 depth_rank=8 area_rank=2
   Limited clear space immediately around the basketball game.
4. basketball player [left-mid] conf=0.36 depth=0.552 depth_rank=3 area_rank=6
   Limited clear space immediately around the basketball player.
5. basketball coach [left-mid] conf=0.35 depth=0.618 depth_rank=5 area_rank=5
   Limited clear space immediately around the basketball coach.
6. crowd [center-far] conf=0.30 depth=0.983 depth_rank=9 area_rank=3
   Limited clear space immediately around the crowd.
7. floor [right-far] conf=0.36 depth=0.699 depth_rank=6 area_rank=9
   Clear space above of the floor.
8. player man [center-near] conf=0.32 depth=0.192 depth_rank=1 area_rank=8
   Limited clear space immediately around the player man.
9. player [left-mid] conf=0.33 depth=0.552 depth_rank=2 area_rank=7
   Limited clear space immediately around the player.

Scene Type A — Grounded spatial description

Based on the scene graph and visual analysis, a "player man" (item 8) is seated in the near-left portion of the image. A standing "basketball coach man" (item 1) is positioned just behind and to the right of the seated man, also in the left-mid foreground. The "basketball coach" (item 5) likely refers to this same standing figure. Further back, in the mid-ground, a group identified as the "basketball team" (item 2) sits on a bench. Individual players, such as the "basketball player" (item 4) and "player" (item 9), are part of this team group. The team faces forward, with a large "crowd" (item 6) filling the stadium seating in the center-far background behind them. The "floor" (item 7) of the basketball court is visible in the bottom-right corner, and the entire setting represents a "basketball game" (item 3) in progress.

Scene Type B — Grounded referring expressions

Object: basketball coach man

Referring: The man standing in the left-mid area, wearing a black hoodie and blue jeans, with his back mostly turned to the camera.
Free space: Limited clear space immediately surrounds the man, as he is standing close to seated individuals and in front of the crowd.

Object: basketball team
Referring: The group of players and staff sitting on the bench in the center of the image.
Free space: Limited clear space immediately surrounds the group, as they are sitting close together on the bench.

Object: basketball game
Referring: The entire scene depicted in the image, showing the players and crowd during a basketball event.
Free space: Limited clear space exists throughout the scene due to the large number of people in the stadium setting.

Object: basketball player
Referring: The player sitting on the far left of the bench group, wearing a green jersey with the number 4.
Free space: Limited clear space immediately surrounds the player, as he is seated between other individuals.

Object: basketball coach
Referring: The man standing in the left-mid area, wearing a black hoodie and blue jeans, with his back mostly turned to the camera.
Free space: Limited clear space immediately surrounds the man, as he is standing close to seated individuals and in front of the crowd.

Object: crowd
Referring: The large group of spectators filling the tiered seating in the background behind the players.
Free space: Limited clear space immediately surrounds individual members of the crowd, as they are seated closely together.

Object: floor
Referring: The blue surface of the basketball court in the foreground where the players and staff are sitting.
Free space: There is limited clear space on the floor itself, as it is occupied by the people and items in the foreground.

Object: player man
Referring: The man sitting in the immediate foreground on the left side, wearing a light blue jacket.
Free space: Limited clear space immediately surrounds the man, as he is sitting close to other individuals.

Object: player
Referring: The player sitting on the far left of the bench group, wearing a green jersey with the number 4.
Free space: Limited clear space immediately surrounds the player, as he is seated between other individuals.

000019.jpg

Original caption

rihanna walks into the airport with her family

Scene graph

SCENE GRAPH (14 objects detected):
1. office building [center-mid] conf=0.54 depth=0.368 depth_rank=8 area_rank=2
   Limited clear space immediately around the office building.
2. person man skateboarder [right-mid] conf=0.38 depth=0.480 depth_rank=13 area_rank=11
   Limited clear space immediately around the person man skateboarder.
3. person skateboarder [center-near] conf=0.44 depth=0.272 depth_rank=4 area_rank=10
   Limited clear space immediately around the person skateboarder.
4. glass door [right-near] conf=0.44 depth=0.091 depth_rank=3 area_rank=3
   Limited clear space immediately around the glass door.
5. skateboarder [center-mid] conf=0.31 depth=0.368 depth_rank=6 area_rank=13
   Limited clear space immediately around the skateboarder.
6. glass door [left-mid] conf=0.46 depth=0.398 depth_rank=11 area_rank=8
   Limited clear space immediately around the glass door.
7. elevator [left-mid] conf=0.32 depth=0.398 depth_rank=9 area_rank=6
   Limited clear space immediately around the elevator.
8. doorway elevator glass door [right-near] conf=0.30 depth=0.091 depth_rank=2 area_rank=4
   Limited clear space immediately around the doorway elevator glass door.
9. building office [center-mid] conf=0.30 depth=0.368 depth_rank=7 area_rank=1
   Limited clear space immediately around the building office.
10. doorway [right-near] conf=0.33 depth=0.091 depth_rank=1 area_rank=5
   Limited clear space immediately around the doorway.
11. doorway [left-mid] conf=0.33 depth=0.398 depth_rank=10 area_rank=7
   Limited clear space immediately around the doorway.
12. man [right-mid] conf=0.34 depth=0.480 depth_rank=12 area_rank=12
   Limited clear space immediately around the man.
13. man [center-near] conf=0.31 depth=0.272 depth_rank=5 area_rank=9
   Limited clear space immediately around the man.
14. person [center-mid] conf=0.36 depth=0.498 depth_rank=14 area_rank=14
   Limited clear space immediately around the person.

Scene Type A — Grounded spatial description

A large group of people is tightly clustered together in the center-near to center-mid portion of the

scene, positioned at a building entrance. The group includes individuals identified as a person [center-near], man [center-near], person [center-mid], and skateboarder [center-mid] in the immediate vicinity, along with a man [right-mid] and person man skateboarder [right-mid] slightly further to the right. The entire group is framed by structural elements on both sides. On the left, a glass door [left-mid] and doorway [left-mid] define the entrance space, with an associated elevator structure [left-mid] part of the glass frame. On the right side, closer to the viewer, another glass door [right-near] and doorway [right-near] are visible, along with a more complex doorway elevator glass door structure [right-near] which helps frame the right edge of the entrance. The background behind the people consists of the main building structure, identified as a building office [center-mid] and office building [center-mid].

Scene Type C — Grounded action caption

Based on scene analysis, the environment presents a significant challenge for manipulation due to high object density. The primary obstacles are a cluster of individuals, identified as `person skateboarder` [3] (depth rank 4) and `man` [13] (depth rank 5), tightly grouped near the foreground. These individuals are located immediately behind the very close `doorway` structures [10] and [8] (depth ranks 1 and 2). Any potential target object within this group requires a careful high-angle or lateral approach to avoid collision with surrounding `person` and `man` entities. The free space around all detected objects is reported as "Limited," making it difficult to find a clear placement zone. A safe action plan requires navigating through this narrow opening while maintaining awareness of the dynamic human cluster.

000017.png

Original caption

a home decor mood board with white linen intentions

Scene graph

SCENE GRAPH (14 objects detected):
1. curtain [left-near] conf=0.41 depth=0.226 depth_rank=10 area_rank=5
   Limited clear space immediately around the curtain.
2. picture picture frame [center-near] conf=0.38 depth=0.151 depth_rank=5 area_rank=8
   Clear space below of the picture picture frame.
3. picture picture frame [center-near] conf=0.38 depth=0.141 depth_rank=3 area_rank=6
   Limited clear space immediately around the picture picture frame.
4. pillow [right-mid] conf=0.45 depth=0.418 depth_rank=13 area_rank=12
   Limited clear space immediately around the pillow.
5. furniture [left-near] conf=0.46 depth=0.152 depth_rank=6 area_rank=10
   Clear space below of the furniture.
6. curtain [left-near] conf=0.40 depth=0.237 depth_rank=11 area_rank=7
   Limited clear space immediately around the curtain.
7. pillow [left-near] conf=0.45 depth=0.156 depth_rank=7 area_rank=13
   Clear space right of the pillow.
8. carpet [center-near] conf=0.43 depth=0.134 depth_rank=2 area_rank=4
   Clear space right of the carpet.
9. pillow [right-mid] conf=0.35 depth=0.431 depth_rank=14 area_rank=14
   Limited clear space immediately around the pillow.
10. picture picture frame [center-near] conf=0.36 depth=0.165 depth_rank=9 area_rank=11
   Limited clear space immediately around the picture picture frame.
11. dresser [left-near] conf=0.38 depth=0.098 depth_rank=1 area_rank=9
   Limited clear space immediately around the dresser.
12. collage photo [center-near] conf=0.36 depth=0.150 depth_rank=4 area_rank=1
   Limited clear space immediately around the collage photo.
13. living room room [right-mid] conf=0.33 depth=0.391 depth_rank=12 area_rank=2
   Clear space below of the living room room.
14. photo frame [left-near] conf=0.31 depth=0.165 depth_rank=8 area_rank=3
   Limited clear space immediately around the photo frame.

000034.jpg

Original caption

a white sofa is next to a glass table

Scene graph

SCENE GRAPH (13 objects detected):
1. vase [right-mid] conf=0.78 depth=0.420 depth_rank=6 area_rank=13
   Clear space below of the vase.
2. flower [right-mid] conf=0.65 depth=0.417 depth_rank=5 area_rank=8
   Limited clear space immediately around the flower.
3. bedroom room [center-far] conf=0.67 depth=0.859 depth_rank=12 area_rank=1
   Limited clear space immediately around the bedroom room.
4. pillow [left-mid] conf=0.46 depth=0.460 depth_rank=8 area_rank=9
   Limited clear space immediately around the pillow.
5. pillow [center-mid] conf=0.43 depth=0.666 depth_rank=11 area_rank=11
   Clear space right of the pillow.
6. carpet [left-near] conf=0.57 depth=0.131 depth_rank=1 area_rank=6
   Limited clear space immediately around the carpet.
7. side table nightstand [right-mid] conf=0.49 depth=0.395 depth_rank=4 area_rank=7
   Clear space below of the side table nightstand.
8. pillow [left-mid] conf=0.43 depth=0.661 depth_rank=10 area_rank=12
   Limited clear space immediately around the pillow.
9. bed [center-near] conf=0.37 depth=0.273 depth_rank=2 area_rank=3
   Limited clear space immediately around the bed.
10. pillow [left-mid] conf=0.43 depth=0.443 depth_rank=7 area_rank=10
   Clear space below of the pillow.
11. bedding blanket comfort linen [center-near] conf=0.31 depth=0.273 depth_rank=3 area_rank=4
   Limited clear space immediately around the bedding blanket comfort linen.
12. bed [center-mid] conf=0.58 depth=0.569 depth_rank=9 area_rank=5
   Limited clear space immediately around the bed.
13. room [center-far] conf=0.32 depth=0.859 depth_rank=13 area_rank=2
   Limited clear space immediately around the room.

Scene Type A — Grounded spatial description

In a minimalist room setting with bright white walls, two daybeds are arranged in the center. The front daybed, positioned in the near-center foreground, features white bedding and a cluster of pillows at its head, along with an open book resting on the surface. Directly behind it and slightly to the right is a

second daybed, which also includes pillows on its bedding. A white side table stands to the right of the second daybed, holding a vase with flowers. In the left foreground, a dark, circular carpet partially covers the floor.

Scene Type B — Grounded referring expressions

Object: vase
Referring: The clear glass vase on the right side table.
Free space: There is clear space on the surface of the side table surrounding the base of the vase.

Object: flower
Referring: The cluster of orange flowers in the glass vase.
Free space: There is limited clear space immediately around the flowers due to their clustering within the vase.

Object: bedroom room
Referring: The general background area of the room containing the furniture.
Free space: The room contains open space around the furniture items.

Object: pillow
Referring: The white pillow located on the left side of the front bed's pillow arrangement.
Free space: The pillow is closely surrounded by other pillows, limiting free space immediately around it.

Object: pillow
Referring: The white pillow on the back bed, to the right of the gray pillow.
Free space: There is clear space on the bed surface to the right of the pillow.

Object: carpet
Referring: The large, dark circular carpet in the foreground.
Free space: There is clear floor space around the edges of the carpet.

Object: side table nightstand
Referring: The white side table positioned to the right of the back daybed.
Free space: There is clear floor space underneath the side table.

Object: pillow
Referring: The gray pillow positioned behind the white pillows on the front bed.
Free space: The pillow is tightly nestled between the headrest and other pillows, limiting immediate free space.

Object: bed
Referring: The white daybed located in the foreground.
Free space: There is limited free space around the front daybed due to its large size and proximity to the image edges.

Object: pillow
Referring: The rightmost white pillow on the front bed's pillow arrangement.
Free space: There is clear space below the pillow on the bed's surface.

Object: bedding blanket comfort linen
Referring: The white bedding covering the front daybed.
Free space: The bedding covers the entire visible area of the bed, with limited free space around its edges.

Object: bed
Referring: The white daybed located in the middle ground, behind the front daybed.

Free space: There is clear space to the right of the bed, next to the side table.

Object: room
Referring: The background area of the room beyond the main furniture arrangement.
Free space: The room contains open space around the furniture items.

Scene Type C — Grounded action caption

Based on the scene analysis, my primary targets for manipulation are the objects on the side table [right-mid] and the pillows on the bed [left-mid]. The vase [depth=0.420] and flowers [depth=0.417] sit on the side table nightstand [depth=0.395]. To grasp the vase safely, a side approach is necessary to avoid disturbing the flowers, which have "limited clear space immediately around" them. Similarly, the pillows on the bed are tightly clustered; grasping any individual pillow requires a precise, careful approach due to the "limited clear space" between adjacent items. For placement tasks, the "clear space right of the pillow" on the front bed [center-mid] offers an accessible and relatively large, flat surface for depositing objects.

000001.jpg

Original caption

the view of an aerial pool with palm trees

Scene graph

SCENE GRAPH (17 objects detected):
1. hotel [left-far] conf=0.55 depth=0.748 depth_rank=16 area_rank=3
   Limited clear space immediately around the hotel.
2. person [right-mid] conf=0.51 depth=0.609 depth_rank=11 area_rank=15
   Clear space above of the person.
3. sea [right-far] conf=0.59 depth=0.899 depth_rank=17 area_rank=4
   Clear space above of the sea.
4. person [center-mid] conf=0.49 depth=0.455 depth_rank=6 area_rank=14
   Clear space left, right, above of the person.
5. person [right-near] conf=0.58 depth=0.033 depth_rank=1 area_rank=6
   Clear space above of the person.
6. person [right-mid] conf=0.51 depth=0.633 depth_rank=13 area_rank=13
   Clear space above of the person.
7. person [right-mid] conf=0.65 depth=0.452 depth_rank=4 area_rank=7
   Limited clear space immediately around the person.
8. person [right-far] conf=0.46 depth=0.686 depth_rank=14 area_rank=16
   Clear space above of the person.
9. person [left-mid] conf=0.45 depth=0.451 depth_rank=3 area_rank=8
   Limited clear space immediately around the person.
10. person [left-mid] conf=0.46 depth=0.524 depth_rank=9 area_rank=10
   Limited clear space immediately around the person.
11. person [left-mid] conf=0.47 depth=0.518 depth_rank=8 area_rank=11
   Clear space below of the person.
12. person [left-near] conf=0.51 depth=0.290 depth_rank=2 area_rank=9
   Clear space above of the person.
13. pool [center-mid] conf=0.68 depth=0.491 depth_rank=7 area_rank=5
   Limited clear space immediately around the pool.
14. person [left-mid] conf=0.37 depth=0.453 depth_rank=5 area_rank=12
   Clear space above of the person.
15. person [left-mid] conf=0.31 depth=0.629 depth_rank=12 area_rank=17
   Clear space left, right, above, below of the person.
16. hotel resort [left-mid] conf=0.35 depth=0.571 depth_rank=10 area_rank=1

Limited clear space immediately around the hotel resort.
17. resort [left-far] conf=0.35 depth=0.748 depth_rank=15 area_rank=2
Limited clear space immediately around the resort.

Scene Type A — Grounded spatial description

Based on the scene graph and visual analysis, the scene centers on a large hotel resort complex (1, 16, 17) situated prominently on the left side of the image, extending from the midground to the far background. Below this structure, in the center-mid area, lies a large pool (13). The sea (3) fills the background on the right side of the image. Multiple people populate the scene at varying distances. In the immediate foreground, a person (5) is visible on the right, while another person (12) walks on a path on the left. Several individuals (9, 10, 11, 14, 15) are situated in the midground near the hotel complex on the left, with some on an upper terrace and others on the lower level near the pool. In the center-mid area, a person (4) is located within the pool. Further away, on the right side, several people (2, 6, 7, 8) walk along the path between the pool area and the sea.

Scene Type B — Grounded referring expressions

Object: hotel
Referring: The large, multi-story building located on the far left side of the image, set into the hillside.
Free space: There is limited clear space immediately around the hotel due to surrounding structures and trees.

Object: person
Referring: The middle person in the group of three people walking along the path near the water's edge on the right side.
Free space: There is clear space above the person and immediately around them on the path.

Object: sea
Referring: The large body of water extending from the mid-ground to the far horizon on the right side of the image.
Free space: There is a clear view of the sky above the sea.

Object: person
Referring: The person standing near the edge of the elevated deck and smaller pool on the left side, wearing a white top.
Free space: There is clear space immediately around the person, especially to the left, right, and above.

Object: person
Referring: The person sitting at a dining table in the very foreground on the bottom right side of the image, facing away from the viewer.
Free space: There is clear space above the person.

Object: person
Referring: The leftmost person in the group of three people walking along the path by the water on the right side of the image.
Free space: There is clear space above the person.

Object: person
Referring: The person standing near the large palm tree to the right of the swimming pool area.
Free space: There is limited clear space immediately around this person due to nearby trees and pool furniture.

Object: person
Referring: The rightmost person in the group of three people walking along the path by the water on the right side of the image.

Free space: There is clear space above the person.

Object: person
Referring: One of the two people standing closely together on the elevated deck on the left side, near the edge overlooking the pool.
Free space: There is limited clear space immediately around the person, as they are standing close to another person and a railing.

Object: person
Referring: One of the two people walking down a ramp near the base of the main building on the left side.
Free space: There is limited clear space immediately around the person due to being on a pathway next to other elements.

Object: person
Referring: The second person walking down the ramp near the base of the main building on the left side, accompanying the other person.
Free space: There is clear ground space below the person on the pathway.

Object: person
Referring: The person standing near a planter in the bottom-left corner of the image.
Free space: There is clear space above the person.

Object: pool
Referring: The large, turquoise-colored swimming pool complex in the center of the image, surrounded by a deck and lounge chairs.
Free space: There is limited clear space immediately around the pool due to lounge chairs, umbrellas, and surrounding foliage.

Object: person
Referring: The second of the two people standing closely together on the elevated deck on the left side.
Free space: There is clear space above the person.

Object: person
Referring: The single person walking on the grassy area on the left side, between the main building and the pool area.
Free space: There is clear space all around the person.

Object: hotel resort
Referring: The large building and associated deck structures forming the resort complex on the left side of the image.
Free space: There is limited clear space immediately around the hotel resort due to surrounding hillside, trees, and other structures.

Object: resort
Referring: The main multi-story building and complex located in the background on the far left side.
Free space: There is limited clear space immediately around the resort complex due to surrounding vegetation and the hillside.